

System Requirement Document of ML Based Agriculture Robot

Title	ML Based Agriculture Robot
Document	System Requirement Documents
Document Ref	RKA_MARS_2V00_SyRD_0V01
Date	22/08/2026
Prepared By	Kamalesh R
Guided By	Banumathi S
Submitted to	MSME

Table of Contents

1 REQUIREMENT.....	3
1.1 Hardware Requirement:.....	3
1.2 Software Requirement:.....	3
1.3 Mechanical Requirement:.....	4
1.4 Power Supply Requirement:.....	4
1.5 External Interface Requirement:.....	5
1.6 Internal Interface Requirement:.....	5
2 SYSTEM SPECIFICATION.....	6

1 REQUIREMENT

1.1 Hardware Requirement:

Requirement ID	Parameter	Requirement
HWR-REQ-1	Category	Commercial - Proto
HWR-REQ-2	Power Supply Source	Internal
HWR-REQ-3	Operating Voltage	12VDC and 7.2VDC
HWR-REQ-4	Current Consumption	Less than 2A @12VDC and 5A @ 7.2V
HWR-REQ-5	External Communication	Bluetooth
HWR-REQ-6	Inter-Module Communication	UART (Modified)
HWR-REQ-7	Navigation	GPS / CV
HWR-REQ-8	Controlling Element	ARM-M0 or higher
HWR-REQ-9	Processing Element	ARM-A30 or higher
HWR-REQ-10	Operating Temperature	0 to +55 deg
HWR-REQ-11	Motor Driver	Should be control direction and speed of 4 DC motor
HWR-REQ-12	Servo Driver	Should be control direction and position of 5 servo motor
HWR-REQ-13	Camera	HD or higher
HWR-REQ-14	Cooling	Air cooling / forced Air cooling

1.2 Software Requirement:

Requirement ID	Parameter	Description
SWR-REQ-1	Processing Language	Python-3.11 or Advanced
SWR-REQ-2	Controlling Language	Micro-Python
SWR-REQ-3	ML/DL Algorithm	CNN
SWR-REQ-4	ML/DL Model	YOLOv4 or Higher
SWR-REQ-5	Inter-module Control Data flow	Host device should control with command through the modified UART interface
SWR-REQ-6	User Control Data Flow	User should be control Robot at runtime with basic command
SWR-REQ-7	Navigation	Based on GPS and CV data, robot perform the movement
SWR-REQ-8	Task Configuration	Task should be configured through HMI using display and keyboard
SWR-REQ-9	Real world Info loading	User should load information Field, type of crop on field through HMI using display and keyboard
SWR-REQ-10	Hardware Control	Speed and position of DC and Servo motors should controllable.

1.3 Mechanical Requirement:

Requirement ID	Parameter	Description
MEC-REQ-1	Dimension	450 x 600 x 720 mm
MEC-REQ-2	Weight	6Kg or less
MEC-REQ-3	Functional Module	Functional block should changeable based on task
MEC-REQ-4	Robot type	Rover
MEC-REQ-5	Main Frame Material	Aluminium
MEC-REQ-6	Movement	Should be movable on rugged surface
MEC-REQ-7	Type of Drive	Wheel
MEC-REQ-8	Driving Source	4x DC motor
MEC-REQ-9	Steering	4x Servo motor
MEC-REQ-10	Functional Tools	Sprayer, Seed picker, Land plougher

1.4 Power Supply Requirement:

Requirement ID	Parameter	Description	
PWS-REQ-1	Converter Isolation Type	Non-Isolated	
PWS-REQ-2	Supply Source type	Two Internal source	
PWS-REQ-3	Supply Source	Li-ion battery	Li-Po battery
PWS-REQ-4	Input Supply Nature	DC	DC
PWS-REQ-5	Input Supply Voltage	7.4V	12V
PWS-REQ-6	Input Current consumption	5A (max)	2A (max)
PWS-REQ-7	Output Supply Nature	DC	DC
PWS-REQ-8	Output Supply Voltage	5V	12V, 5V, 3.3V
PWS-REQ-9	Output Current consumption	5A (max)	1A, 1A, 0.5A (max)
PWS-REQ-10	Charing Voltage	8.5 V	12 - 14V
PWS-REQ-11	Battery Capacity	75Wh	50Wh
PWS-REQ-12	On/Off Control	SPST switch	SPST switch
PWS-REQ-13	Power Consumption	40 W	
PWS-REQ-14	Endurance	1.5 Hrs min	
PWS-REQ-15	Protection	BMS should be used for OC, UV protection.	

1.5 External Interface Requirement:

Requirement ID	Parameter	Description
EXI-REQ-1	Li-ion Battery Connector	2 pin Jumper
EXI-REQ-2	Li-Po Battery Connector	2 pin Solder points
EXI-REQ-3	Charging Connector	2 Nos of 2-pin round connector to charge batteries
EXI-REQ-4	Function Module Connector	6 Pin connector to interface the functional module
EXI-REQ-5	HMI	USB port for interfacing Keyboard and mouse

1.6 Internal Interface Requirement:

Requirement ID	Parameter	Description
INI-REQ-1	HOST-COM connector	4x10 pins for power and communication to modules
INI-REQ-2	DS-COM connector	5 pins for power and communication to HOST modules
INI-REQ-3	DS-MOT connectors ⁽¹⁾	4 Nos of 2 terminal connector for driving motor
INI-REQ-4	DS-SERVO connector	4 Nos of 3 pin connectors for steering motor
INI-REQ-5	PI-HDMI ⁽¹⁾	Micro HDMI port for interfacing display
INI-REQ-6	PI-USB ⁽¹⁾	USB-A for Camera Interface and power the display USB-C for Input Power supply of PI from HAT
INI-REQ-7	PI-GPIO ⁽¹⁾	20x2 pins connector for COM from HOST
INI-REQ-8	HOST-PWR connector	3 Nos of 3 terminal connector block for 5 and 12 supply
INI-REQ-9	MOTOR-CON connector ⁽¹⁾	2 solder point of DC motor
INI-REQ-10	SERVO-CON connector ⁽¹⁾	3 pin connector Servo motor
INI-REQ-11	Camera Interface ⁽¹⁾	USB-A connector to PI
INI-REQ-12	DISPLAY-HDMI ⁽¹⁾	HDMI port to interface PI
INI-REQ-13	DISPLAY-USB ⁽¹⁾	Micro-USB for Input power supply

Note: ⁽¹⁾ Connector is part of Component

2 SYSTEM SPECIFICATION

Sl.No	Parameter	Specification
1	Category	Commercial - Proto
2	Stage	Proto
3	Power Supply Source	Internal Battery
4	Operating Voltage	12VDC and 7.2VDC
5	Current Consumption	2A (max) @12VDC and 5A (Max) @ 7.2V
6	Dimension	600 x 450 x 700 mm
7	Weight	Less than 6Kg
8	External Communication	Bluetooth
9	Inter-Module Communication	UART (Modified)
10	Control	Automatic / Semi-Automatic
11	Navigation	GPS / CV
12	ML/DL Model	YOLOv4 or advanced
13	Controlling Element	RP2040 (PICO)
14	Processing Element	Raspberry Pi-5